Ngày.....Tháng.....Năm.....

Đề 2013-2014 chuẩn Anh

Câu 1:

a) $h(n) = (h_1(n) * h_2(n)) + (h_3(n) * h_4(n))$

$h_1(n) * h_2(n) = \{-1, -1, 0, -1, 1\}$

	1	1	-1
→ -1	-1	-1	1
0	0	0	0
-1	-1	-1	1

	1	0	-1
2	2	0	-2
→ 0	0	0	0
-1	-1	0	1

$h_3(n) * h_4(n) = \{2, 0, -3, 0, 1\}$

$h(n) = \{2, 0, -4, -1, 1, -1, 1\}$

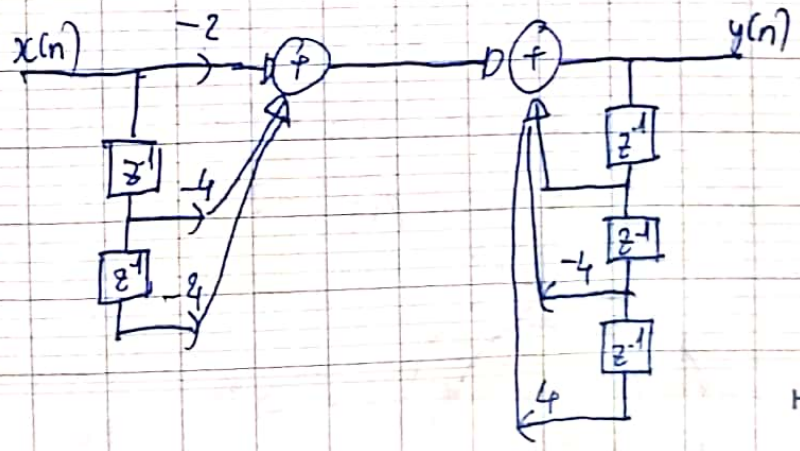
b) $y(n) = x(n) * h(n)$

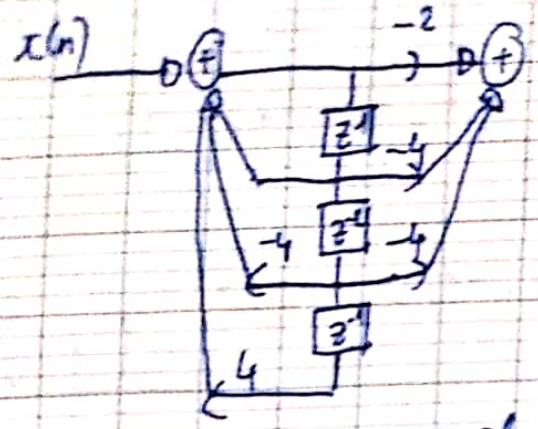
	2	0	-4	-1	1	-1	1
→ 0	0	0	0	0	0	0	0
-1	-2	0	4	1	-1	1	-1
2	4	0	-8	-2	2	-2	2

$y(n) = \{0, -2, 4, 4, -7, -3, 3, -3, 2\}$

Câu 2: $y(n) = -2x(n) - 4x(n-1) - 4x(n-2) + y(n-1) - 4y(n-2) + 4y(n-3)$

a)





b) $Y(z) = -2X(z) - 4X(z).z^{-1} - 4X(z).z^{-2} + Y(z).z^{-1} - 4Y(z).z^{-2} + 4Y(z).z^{-3}$

$\Rightarrow Y(z)\{1 - z^{-1} + 4z^{-2} - 4z^{-3}) = X(z)(-2 - 4z^{-1} - 4z^{-2})$

$\Rightarrow \dfrac{Y(z)}{X(z)} = \dfrac{-2 - 4z^{-1} - 4z^{-2}}{1 - z^{-1} + 4z^{-2} - 4z^{-3}} = H(z).$

$\Rightarrow H(z) = \dfrac{-2 - 4z^{-1} - 4z^{-2}}{(1-z^{-1})(1-2jz^{-1})(1+2jz^{-1})}$

$= \dfrac{A}{1-z^{-1}} + \dfrac{B}{1-2jz^{-1}} + \dfrac{B^*}{1+2jz^{-1}}$

$A = \lim\limits_{z^{-1}\to 1} (1-z^{-1})H(z) = \dfrac{-2-4-4}{(1-2i)(1+2i)} = -2.$

$B = \lim\limits_{z^{-1}\to \frac{1}{2i}} (1-2iz^{-1}).H(z) = \dfrac{-2-4.\frac{1}{2i} - 4(\frac{1}{2i})^2}{(1-\frac{1}{2i})(1+\frac{2i}{2i})} = e^{j90^\circ} = e^{j\pi/2}$

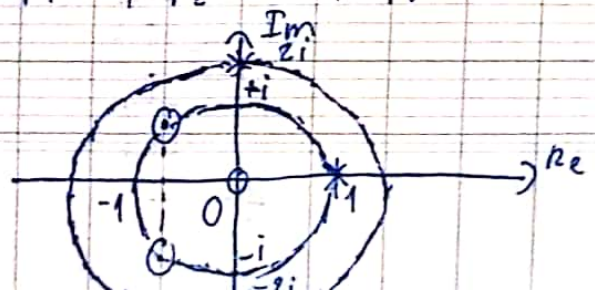
$B^* = e^{-j90^\circ}$

$\Rightarrow h(n) = -2u(n)(1)^n + e^{j\pi/2}(+2i)^n u(n) + e^{-j\pi/2}(-2i)^n u(n)$

c) $H(z) = \dfrac{-2z^3 - 4z^2 - 4z}{(z-1)(z-2i)(z+2i)} = \dfrac{z(z+(1-i))(z+(1+i))}{(z-1)(z-2i)(z+2i)}$

$z_1 = 0,\ z_2 = -1+i,\ z_3 = -1-i$

$p_1 = 1,\ p_2 = 2i,\ p_3 = -2i$



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Câu 3: $y(n) = 2x(n) + 2x(n-1) + y(n-1)$

a) $Y(\Omega) = 2X(\Omega) + 2e^{-j\Omega}X(\Omega) + e^{-j\Omega}Y(\Omega)$

$\rightarrow Y(\Omega)(1-e^{-j\Omega}) = X(\Omega)(2+2e^{-j\Omega})$

$$\rightarrow \frac{Y(\Omega)}{X(\Omega)} = \frac{2+2e^{-j\Omega}}{1-e^{-j\Omega}} = H(\Omega)$$

$$= \frac{2(1+e^{-j\Omega})}{1-e^{-j\Omega}} = \frac{4\cos(\Omega/2)e^{-j\frac{\Omega}{2}}}{2j\sin(\frac{\Omega}{2})e^{-j\frac{\Omega}{2}}} = \frac{4\cos(\Omega/2)}{2j\sin(\Omega/2)}$$

b) $|H(\Omega)| = \left|\frac{4\cos(\Omega/2)}{2j\sin(\Omega/2)}\right|$ (Có biết vẽ ∧.∧)

$$\varphi(\Omega) = \arg\left[\frac{4\cos(\Omega/2)}{2j\sin(\Omega/2)}\right]$$

c) $x(n) = e^{jn\pi/2} \longrightarrow \Omega = \pi/2$

$|X(\frac{\pi}{2})| = 1$

$\varphi_X(\frac{\pi}{2}) = 0°$

$$H(\Omega) = \frac{2+2e^{-j\Omega}}{1-e^{-j\Omega}}$$

$|Y(\Omega)| = |X(\Omega)|\,|H(\Omega)|$

$\varphi_Y(\Omega) = \varphi_X(\Omega) + \varphi_H(\Omega)$

$\Rightarrow |Y(\frac{\pi}{2})| = |X(\frac{\pi}{2})|.|H(\frac{\pi}{2})|$ $\quad H(\frac{\pi}{2}) = \frac{2+2e^{-j\pi/2}}{1-e^{-j\pi/2}} = 2\angle 90°$

$= 1 \cdot 2 = 2$

$\varphi_Y(\frac{\pi}{2}) = 0 + 90° = 90°$

$\Rightarrow y(n) = 2.e^{j(n\frac{\pi}{2} + 90°)}$

Câu 4: a) Chập vòng tuần hoàn $x(n) = \{\underset{\uparrow}{0}, 1, 2, 3\}$

$y(n) = x(n) \underset{4}{\otimes} x(n)$

$\rightarrow v(n) = x(n) * x(n) = \{\underset{\downarrow}{0}, 0, 1, 4, 10, 12, 9\}$

	$\underset{\downarrow}{0}$	1	2	3
$\rightarrow 0$	0	0	0	0
1	0	1	2	3
2	0	2	4	6
3	0	3	6	9

$y(n) = v(n) + v(n-4) + v(n+4)$

$v(n-4) = \{\underset{\downarrow}{0}, 0, 0, 0, 0, 0, 1\}$

$v(n+4) = \{\underset{\downarrow}{10}, 12, 9, 0, 0, 0, 0\}$

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b) DFT 8 chiều $x(n) = \{\overset{\downarrow}{0}, 1, 2, 3\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) . e^{-j2\pi kn/N} \qquad N = 8$$

$$= x(0) . e^{-j2\pi k.0/8} + x(1) . e^{-j2\pi k/8} + x(2) e^{-j4\pi k/8} + x(3) . e^{-j6\pi k}$$

$$= e^{-j\pi k/4} + 2e^{-j\pi k/2} + 3.e^{-j3\pi k/4} \qquad 0 \leq k \leq 7$$

c) $x(n) * y(n) \overset{DT}{\underset{FT}{\longleftrightarrow}} X(\Omega) Y(\Omega)$

$$X(\Omega) = \sum_{n=-\infty}^{+\infty} x(n) e^{-jn\Omega}$$

$$X(\Omega).Y(\Omega) = \sum_{n=-\infty}^{+\infty} [x(n) * y(n)] . e^{-jn\Omega}$$

$$= \sum_{n=-\infty}^{+\infty} \left[\sum_{k=-\infty}^{+\infty} x(k) y(n-k) \right] . e^{-jn\Omega}$$

Đặt $n - k = m$

$n = m + k$

$$\Rightarrow \sum_{k=-\infty}^{+\infty} x(k) \sum_{m+k=-\infty}^{+\infty} y(m) e^{-j\Omega(m+k)}$$

$$\Rightarrow \sum_{k=-\infty}^{+\infty} x(k) \sum_{m=-\infty}^{+\infty} y(m) e^{-j\Omega m} . e^{-j\Omega k}$$

$$\Rightarrow \sum_{k=-\infty}^{+\infty} x(k) e^{-j\Omega k} . \sum_{m=-\infty}^{+\infty} y(m) e^{-j\Omega m}$$

$$\Rightarrow X(\Omega) . Y(\Omega)$$